Day 111: The Week 16 Build — Trees Against Neural Networks on Tabular Data

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 16, Day 7 | Complete study guide

Watch: Why Deep Neural Networks (DNNs) Underperform Tree-Based Models on Tabular Data (DataMListic) | Why do tree based models still outperform deep learning on tabular data? (AI Paper Cast)

Script: `day111_trees_vs_neural_nets.py` · every number in this guide is copied by program from `day111_run_output.txt`, the real output of that script.

The week's six exercise parts, copied in verbatim, become four tree models, and they face a neural network written today. There is no single winner. Trees lead on Titanic (0.8213 against 0.8084 for the network), the network leads on breast cancer (0.9690 against 0.9596), and a plain linear model beats both there (0.9749) and on Diabetes. The more useful results are about *why*: the network is hurt badly by useless columns and by squashed features, and trees are hurt by rotated ones, which matches a well-known 2022 study of the question (Grinsztajn, Oyallon and Varoquaux).

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

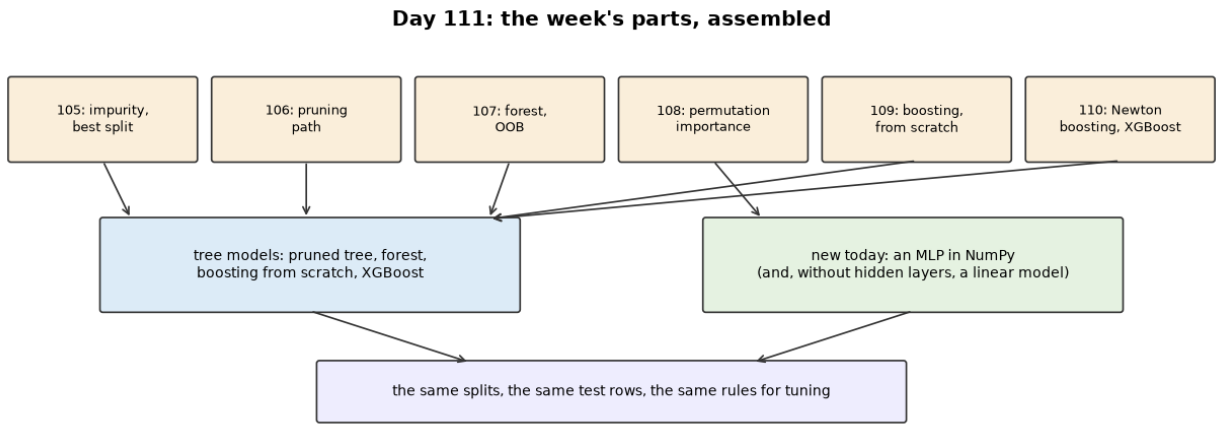
Tabular data is a table of mixed measurements: one row per case, columns in different units, such as age, fare and class. Trees ask one column at a time; a **multilayer perceptron** (MLP) mixes all columns in every unit of every layer.

Day 111 is the week's **build**: the exercise parts of Days 105 to 110 are copied in unchanged and assembled, and a new MLP is written to face them on the same splits.

REAL-WORLD SCENARIO

A loan officer's checklist (income above X? missed payments?) against a scoring formula that blends everything at once. Each suits some applicants better, and the interesting question is which kinds of data favour which.

SYSTEM DIAGRAM



The six exercise parts, the four tree models they build, and the new MLP, all judged on the same splits.

Data: Titanic (the repository's own copy), and breast cancer, Diabetes and digits, bundled with scikit-learn. Everything is NumPy except Day 110's part, which wraps the XGBoost library.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. Checks: the parts are verbatim, and the MLP's gradients are right

DEFINITION — start here, no prior knowledge assumed

The script reads every function in the six exercise files and checks that each appears in itself character for character. The MLP's backpropagation is compared with finite differences: nudge each weight both ways and measure the change in loss.

REAL-WORLD SCENARIO

Checking that the parts on the assembly line are exactly the parts that passed inspection.

IMPLEMENTATION

```
class MLP:
    def __init__(self, sizes, task, seed):
        rng = np.random.RandomState(seed)
        self.task = task
        self.W = [rng.randn(a, b) * np.sqrt(2.0 / a) for a, b in zip(sizes[:-1], sizes[1:])]
        self.b = [np.zeros(b) for b in sizes[1:]]
        self.rng = rng
    def forward(self, X):
        self.A = [X]
        for i, (W, b) in enumerate(zip(self.W, self.b)):
            Z = self.A[-1] @ W + b
            self.A.append(np.maximum(Z, 0) if i < len(self.W) - 1 else Z)
        return self.A[-1]
    def loss_grad(self, Z, y):
        n = len(Z)
        if self.task == "binary":
            return float(np.mean(np.logaddexp(0, Z[:, 0]) - y * Z[:, 0])), (0.5 * (1 + np.tanh(Z /
                2)) - y[:, None]) / n
        if self.task == "multi":
            Z = Z - Z.max(1, keepdims=True); P = np.exp(Z); P /= P.sum(1, keepdims=True)
            Y = np.eye(Z.shape[1])[y]
            return float(-np.mean(np.log(P[np.arange(n), y] + 1e-12))), (P - Y) / n
        return float(np.mean((Z[:, 0] - y) ** 2) / 2), (Z - y[:, None]) / n
    def grads(self, X, y):
        loss, dZ = self.loss_grad(self.forward(X), y)
        gW, gb = [None] * len(self.W), [None] * len(self.W)
        for i in range(len(self.W) - 1, -1, -1):
            gW[i] = self.A[i].T @ dZ; gb[i] = dZ.sum(0)
            if i:
                dZ = (dZ @ self.W[i].T) * (self.A[i] > 0)
        return loss, gW, gb
```

ACTUAL OUTPUT

```
functions from day105_exercise.py to day110_exercise.py found here, character for character: 34 of
    34
MLP gradient check, binary    : largest gap between backpropagation and finite differences 1.1e-10
MLP gradient check, multi     : largest gap between backpropagation and finite differences 2.3e-10
MLP gradient check, regression: largest gap between backpropagation and finite differences 1.1e-09
Actual output from this lesson's run.
```

What it means

All 34 of 34 functions are present unchanged, and the gradients agree to about 10^{-10} for all three tasks (1.1e-10, 2.3e-10, 1.1e-09). The network is the same code for classification and regression; without hidden layers it is logistic or linear regression, which gives the baseline for free.

WHY CHOOSE THIS?

The build cannot quietly drift from the parts it claims to use.

WHY NOT THIS? (TRADEOFFS)

A verbatim check proves the copy, not that the parts suit this problem.

2. The head-to-head

DEFINITION — start here, no prior knowledge assumed

Six models on four problems, the same splits for all. Each tunes itself on its own training part only: the pruned tree by cross-validation, the forest's leaf size by out-of-bag accuracy, both boosters by early stopping, and the MLP over 4 settings on a validation part.

REAL-WORLD SCENARIO

A tournament where every player warms up on the same pitch and plays the same matches.

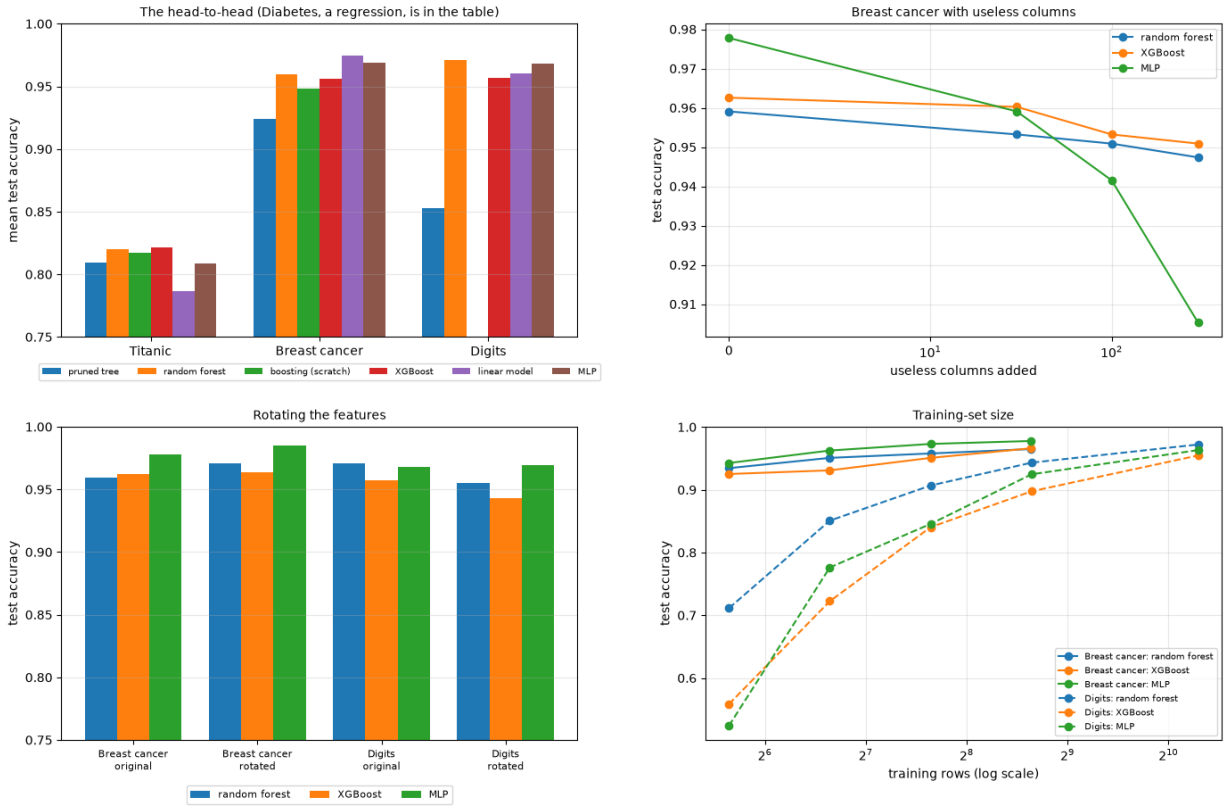
SPECIFICATION

Problems	Titanic (891 rows, 6 features), breast cancer (569, 30), Diabetes (442, 10, regression), digits (1,797, 64).
MLP	Two hidden layers of 32 or 128 ReLU units, weight decay 0.0001 or 0.1, Adam, early stopping, inputs standardised.
Note	The intervals compare the MLP with the best tree model on each problem; the splits share rows, so they are too narrow.

ACTUAL OUTPUT

```
Titanic: 891 rows, 6 features, 20 splits
  model                accuracy     std  seconds
  pruned tree            0.8093  0.0253     0.26
  random forest          0.8201  0.0186     2.98
  boosting (scratch)     0.8174  0.0197     0.64
  XGBoost                0.8213  0.0198     0.02
  linear model           0.7864  0.0273     0.30
  MLP                    0.8084  0.0231     0.62
  MLP minus the best tree model (XGBoost): -0.0129 [-0.0218, -0.0049]; MLP better on 5, worse on 13 of 20
Breast cancer: 569 rows, 30 features, 20 splits
  model                accuracy     std  seconds
  pruned tree            0.9243  0.0147     0.09
  random forest          0.9596  0.0128     0.89
  boosting (scratch)     0.9485  0.0187     1.36
  XGBoost                0.9564  0.0174     0.04
  linear model           0.9749  0.0127     0.13
  MLP                    0.9690  0.0114     0.34
  MLP minus the best tree model (random forest): +0.0094 [+0.0032, +0.0155]; MLP better on 11, worse on 3 of
      20
```

```
Diabetes: 442 rows, 10 features, 20 splits
  model                      RMSE     std  seconds
  boosting (scratch)      57.9726  3.1239     0.66
  XGBoost                 57.7253  3.1854     0.01
  linear model            56.5429  3.3378     0.11
  MLP                     59.6204  2.9229     0.22
  MLP minus the best tree model (XGBoost): +1.8951 [+0.3162, +3.4412]; MLP better on 7, worse on 13 of 20
Digits: 1797 rows, 64 features, 5 splits
  model                 accuracy     std  seconds
  pruned tree             0.8531  0.0194     1.42
  random forest           0.9707  0.0058    14.48
  XGBoost                 0.9570  0.0097     0.60
  linear model            0.9603  0.0049     0.68
  MLP                     0.9681  0.0061     3.22
  MLP minus the best tree model (random forest): -0.0026 [-0.0096, +0.0037]; MLP better on 2, worse on 3 of 5
the splits share rows, so these intervals are too narrow, a rough guide only; seconds are medians per fit,
    within this run
Actual output from this lesson's run.
```



Top left: the head-to-head. Top right: breast cancer with useless columns. Bottom left: rotated features. Bottom right: training-set size.

What it means

The scoreboard is split. On Titanic trees win: XGBoost 0.8213 against the MLP's 0.8084, worse on 13 of 20 splits. On breast cancer the MLP beats the best tree model (+0.0094), but the *linear* model beats everything (0.9749). On Diabetes the linear model is again best (56.5429 RMSE) and the MLP worst (59.6204). On digits, the forest and the MLP are level (-0.0026). With tables this small, the simplest adequate model is often the winner, and 'trees always beat neural networks' does not survive the test.

WHY CHOOSE THIS?

Every model gets the same data and the same tuning rules.

WHY NOT THIS? (TRADEOFFS)

Four small tables; the 2022 study used larger ones, where trees won more clearly.

3. Useless columns

DEFINITION — start here, no prior knowledge assumed

Random columns with no relation to the label are added to breast cancer. Trees can simply never ask about them; an MLP's first layer mixes them into every unit.

REAL-WORLD SCENARIO

A hiring panel given a pile of irrelevant paperwork with every application.

ACTUAL OUTPUT

```
5 splits; the columns are random numbers with no relation to the label
                                    random forest        XGBoost            MLP
  0 useless columns (30 in all)             0.9591         0.9626         0.9778
  30 useless columns (60 in all)            0.9532         0.9602         0.9591
  100 useless columns (130 in all)          0.9509         0.9532         0.9415
  300 useless columns (330 in all)          0.9474         0.9509         0.9053
Actual output from this lesson's run.
```

What it means

The MLP starts best (0.9778) and falls fastest: 0.9053 with 300 useless columns, against 0.9474 for the forest and 0.9509 for XGBoost. This is one of the 2022 study's main explanations, reproduced: real tables often carry many columns that do not matter, and trees are far more robust to them.

WHY CHOOSE THIS?

A direct test of one reason trees win on real tables.

WHY NOT THIS? (TRADEOFFS)

Pure noise columns are an extreme case; real useless columns are often partly correlated with useful ones.

4. Rotation

DEFINITION — start here, no prior knowledge assumed

Features are standardised and multiplied by a random orthogonal matrix: every new feature is a mix of all the old ones. The information is identical. Trees split on one axis at a time, so a rotation should hurt them; an MLP mixes features anyway.

REAL-WORLD SCENARIO

A map turned by 37 degrees: a road map is just as accurate, but street-by-street directions ('go north, then east') no longer fit.

ACTUAL OUTPUT

```
5 splits; features standardised with training statistics, then rotated by a random orthogonal
    matrix
(the same information, so a model that does not care about the axes should not change)
                                    random forest        XGBoost            MLP
  Breast cancer: original axes             0.9591         0.9626         0.9778
  Breast cancer: rotated                   0.9708         0.9637         0.9848
  Digits: original axes                    0.9707         0.9570         0.9681
  Digits: rotated                          0.9551         0.9432         0.9692
Actual output from this lesson's run.
```

What it means

On digits, as expected: the forest falls from 0.9707 to 0.9551 and XGBoost from 0.9570 to 0.9432, while the MLP holds (0.9681 to 0.9692). Trees depend on the columns meaning something on their own. Breast cancer went the other way for the forest (0.9591 to 0.9708), perhaps because mixing its many near-duplicate size measurements gives each split more to work with; that is a guess, not tested here, and on 5 splits the gap may be noise.

WHY CHOOSE THIS?

Shows where trees' advantage comes from: meaningful individual columns.

WHY NOT THIS? (TRADEOFFS)

One random rotation per split; a different one could give a different size of effect.

5. Monotone distortions

DEFINITION — start here, no prior knowledge assumed

Each feature is squashed or stretched by a strictly increasing function, here exp(3z), which keeps every order but creates long tails. Trees only use orders, so they should not care; an MLP uses distances, so it should.

REAL-WORLD SCENARIO

Ranking runners by finishing order is unaffected if the clock runs fast; averaging their times is not.

ACTUAL OUTPUT

```
breast cancer, one split: Day 105's tree on raw and on exp(3z) features gives the same
    probabilities on 1.0000 of TRAINING rows
  and on 0.9942 of test rows: thresholds sit halfway between neighbouring training values, and
      halfway is not kept by exp,
  so a test row that falls between two training values can land on the other side
5 splits:
                                    random forest        XGBoost            MLP
  raw features                             0.9591         0.9626         0.9778
  exp(3z): long right tails                0.9591         0.9626         0.7345
  ranks: every feature uniform             0.9532         0.9626         0.9649
Actual output from this lesson's run.
```

What it means

On training rows Day 105's tree is unchanged (1.0000). On test rows it agrees on 0.9942, not 1: a threshold sits halfway between two training values, halfway is not preserved by exp, and so a test row inside that gap can switch sides. The forest and XGBoost are unmoved; the MLP collapses from 0.9778 to 0.7345, despite standardising its inputs, because standardising cannot undo a long tail. Replacing values by their ranks rescues it (0.9649). For a network, preprocessing is part of the model.

WHY CHOOSE THIS?

Trees need almost no preprocessing.

WHY NOT THIS? (TRADEOFFS)

A network can be just as good, but only if someone shapes the inputs first.

6. Training-set size

DEFINITION — start here, no prior knowledge assumed

The same models trained on 50 rows and upwards, tested on the same 30% of each dataset.

REAL-WORLD SCENARIO

A new employee with a week of experience against one with a year.

ACTUAL OUTPUT

```
5 splits; the test part is the usual 30%; training rows drawn from the rest
Breast cancer:
  training rows     random forest        XGBoost            MLP   linear model
  50                       0.9345         0.9251         0.9427         0.9310
  100                      0.9509         0.9310         0.9626         0.9497
  200                      0.9579         0.9509         0.9731         0.9673
  398                      0.9649         0.9661         0.9778         0.9825
Digits:
  training rows     random forest        XGBoost            MLP   linear model
  50                       0.7113         0.5584         0.5239         0.6356
  100                      0.8508         0.7224         0.7763         0.7822
  200                      0.9069         0.8404         0.8456         0.8664
  400                      0.9432         0.8976         0.9247         0.9154
  1257                     0.9722         0.9547         0.9633         0.9581
Actual output from this lesson's run.
```

What it means

No general rule. On breast cancer the MLP leads at every size (0.9427 with 50 rows) and the linear model finishes top with all 398 (0.9825). On digits the forest leads clearly with little data (0.7113 with 50 rows against 0.5239 for the MLP) and is still ahead with all of it. Which model needs less data depends on the problem's shape.

WHY CHOOSE THIS?

Learning curves show what a single score hides.

WHY NOT THIS? (TRADEOFFS)

Five splits at each size; the smallest sizes are noisy.

7. What each model relies on

DEFINITION — start here, no prior knowledge assumed

Day 108's permutation importance, which works for any model, on Titanic with Day 108's two useless columns: PassengerId and a random coin.

REAL-WORLD SCENARIO

Asking three experts which facts they used, by hiding one fact at a time and watching their answers change.

ACTUAL OUTPUT

```
Titanic with Day 108's PassengerId and coin columns; 5 splits; accuracy lost when a column is shuffled
  feature         random forest        XGBoost            MLP
  is_female              0.1871         0.2139         0.1923
  Pclass                 0.0446         0.0851         0.0466
  Age                    0.0381         0.0526         0.0286
  Fare                   0.0143         0.0107         0.0012
  SibSp                  0.0108         0.0073         0.0160
  PassengerId            0.0102         0.0057         0.0040
  Parch                  0.0102        -0.0007         0.0082
  coin                   0.0019        -0.0010         0.0042
rank correlation (Day 108's spearman): forest-XGBoost 1.000, forest-MLP 0.690, XGBoost-MLP 0.690
Actual output from this lesson's run.
```

What it means

All three lean on the same facts: sex first (0.1871, 0.2139, 0.1923), then class and age. The two tree models rank the features identically (1.000); the MLP agrees less (0.690), using fare least of all. None of them leans much on the useless columns on held-out rows. On this question the models differ less than their architectures suggest.

WHY CHOOSE THIS?

A model-agnostic way to compare what different families learned.

WHY NOT THIS? (TRADEOFFS)

Small effects (below about 0.01) are within shuffle noise.

Interview preparation — technical questions

Q: How do you compare models fairly?

A: The same splits and test rows, and each model tuned only on its own training part.

Q: How do you check a neural network's backpropagation?

A: Finite differences: here gaps of about 1e-10.

Q: Why do neural networks need preprocessing on tables?

A: They use distances and magnitudes: exp-distorted features dropped the MLP from 0.9778 to 0.7345.

Q: Which model would you try first on a new table?

A: A linear model and gradient-boosted trees: the linear model won two of four problems here.

Q: How do you compare what two model families rely on?

A: Permutation importance on held-out rows; here rank correlation 0.690 between forest and MLP.

Interview preparation — theoretical questions

Q: Why do trees often beat neural networks on tabular data?

A: Robustness to useless columns and use of meaningful individual columns: both measured here.

Q: Why are trees not rotation-invariant?

A: Each split asks about one original column; rotated digits cost the forest 0.0156.

Q: Why are trees invariant to monotone transforms?

A: They use only orders; the training partition is unchanged (1.0000).

Q: When can a neural network win on tables?

A: With well-shaped inputs and smooth structure: here on breast cancer, by +0.0094 over the best tree.

Key takeaway and what's next

> **KEY TAKEAWAY**
> On these four tables there is no universal winner: trees won on Titanic, the network on breast cancer, a straight line on two problems. The experiments explain why the debate exists. Trees shrug off useless columns and distorted scales, and depend on columns meaning something; networks shrug off rotations, and depend on someone preparing their inputs.

- Titanic: XGBoost 0.8213, MLP 0.8084. Breast cancer: linear 0.9749, MLP 0.9690, forest 0.9596.
- 300 useless columns: MLP 0.9053, forest 0.9474, XGBoost 0.9509.
- Rotated digits: forest 0.9707 to 0.9551; MLP 0.9681 to 0.9692.
- exp(3z) features: MLP 0.7345, trees unchanged; ranks rescue the MLP (0.9649).

Suggested watch order, if starting from zero

- 1. Why Deep Neural Networks (DNNs) Underperform Tree-Based Models on Tabular Data (DataMListic) — the 2022 study's explanations, in brief.
- 2. Why do tree based models still outperform deep learning on tabular data? (AI Paper Cast) — a reading of the same paper.

Day 112 preview

- **Week 17 begins: support vector machines from scratch**, on real linearly separable and non-separable data.